Module 15: Population and Systems Ecology - Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Compare exponential and logistic population growth.
2. Explain carrying capacity and limiting factors.
3. Distinguish density-dependent and density-independent regulation.
4. Use trophic levels and the 10 percent rule to explain energy limits.
5. Contrast energy flow with nutrient cycling.

Topics

- Population growth models
- Limits and carrying capacity
- Community interactions
- Energy flow through trophic levels
- Matter cycling and systems feedback

Key Terms to Know

- **Population** - Members of one species in an area.
- **Exponential growth** - Growth pattern with increasing rate under abundant resources.
- **Logistic growth** - Growth that slows as population approaches carrying capacity.
- **Carrying capacity** - Approximate environment-supported population size.
- **Density-dependent factor** - Limiting factor whose impact changes with crowding.
- **Density-independent factor** - Limiting factor not directly tied to density.
- **Trophic level** - Feeding position in an energy pathway.
- **Decomposer** - Organism that breaks down dead material and recycles nutrients.

Core Contents

1. Population size changes through births, deaths, immigration, and emigration.
2. Exponential growth describes rapid increase; logistic growth includes carrying capacity.
3. Density-dependent and density-independent factors limit populations differently.
4. Energy flows through producers, consumers, and decomposers with loss at transfers.
5. Matter cycles through ecosystems while energy must continually enter.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-15_population-systems-ecology.md`

Generated Visuals

- Module 15: Population and Systems Ecology Concept Map
- Module 15: Population and Systems Ecology Process Model
- Module 15: Population and Systems Ecology Retrieval Card